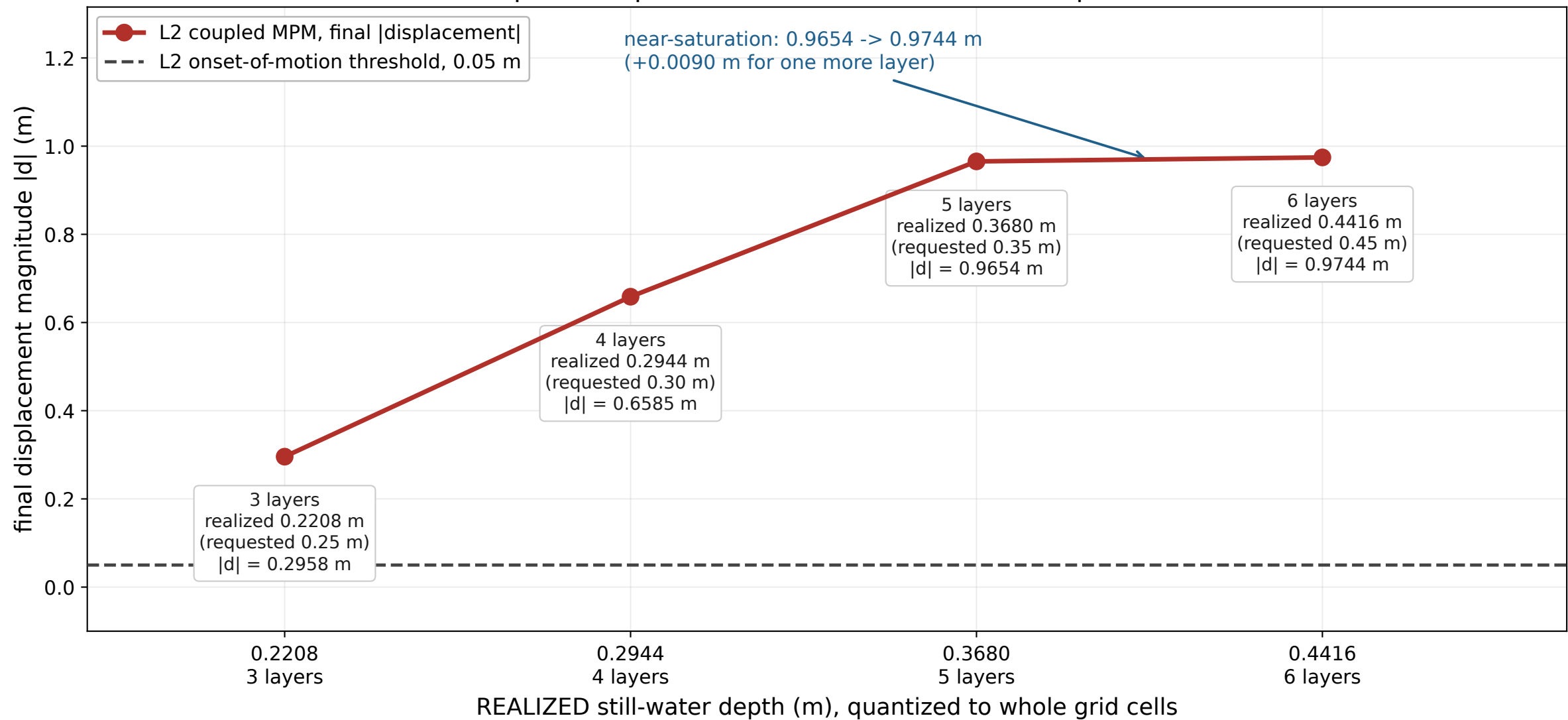


G2 Depth sweep on the realized axis, not the requested one



The water slab is built from whole grid cells, so the requested depth is never the depth simulated. At grid 64 one layer is $h = 0.073607$ m, and the realized depth is $\text{water_layers} \times h$. Requested 0.45 m realizes as 0.4416 m, six layers. Requested 0.25, 0.30 and 0.35 m realize as 0.2208, 0.2944 and 0.3680 m. The requested value is never plotted here.

Displacement rises steeply and then nearly saturates: the last two points are 0.9654 m and 0.9744 m, a gain of only 0.0090 m for a whole extra layer, so the response is flattening at the top of this depth range rather than continuing to climb.

Fixed grid 64, fixed 1.5 m/s surge, one hull at 1100 kg, all runs deterministic. Source: data/all_runs_inventory.csv, runs sweepD_g64_d0p25, g64_m1100, sweepD_g64_d0p35, sweepD_g64_d0p45.